

Linear Algebra – MAT 2610

Section 4.4 (Coordinate Systems)

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Theorem 8

Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for vector space V . Then for each x in V , there exists a unique set of scalars $c_1, \dots, c_n$ such that

$$x = c_1 b_1 + \dots + c_n b_n$$



Theorem 8 - Proof

Since $\mathcal{B}$ spans V , there exist scalars such that $x = c_1 b_1 + \cdots + c_n b_n$ for some scalars $c_1, \dots, c_n$. It remains to show that these are unique.

Suppose there is a second, different set of scalars $d_1, \dots, d_n$ such that

$$x = d_1 b_1 + \cdots + d_n b_n$$

Subtract one expression from the other:

$$(x - x) = d_1 b_1 + \cdots + d_n b_n - (c_1 b_1 + \cdots + c_n b_n)$$

$$0 = (d_1 - c_1)b_1 + \cdots + (d_n - c_n)b_n$$

Some of these coefficients must be non-zero, but this violates $\mathcal{B}$ being a basis.



Definition

Suppose $\mathcal{B} = \{b_1, \dots, b_n\}$ is a basis for V and x is in V . The **coordinates of x relative to the basis $\mathcal{B}$** (or the $\mathcal{B}$ -coordinate of x) are the weights $c_1, \dots, c_n$ such that

$$x = c_1 b_1 + \dots + c_n b_n$$

We say the **coordinate vector of x relative to $\mathcal{B}$** is

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \dots \\ c_n \end{bmatrix}$$



Definition

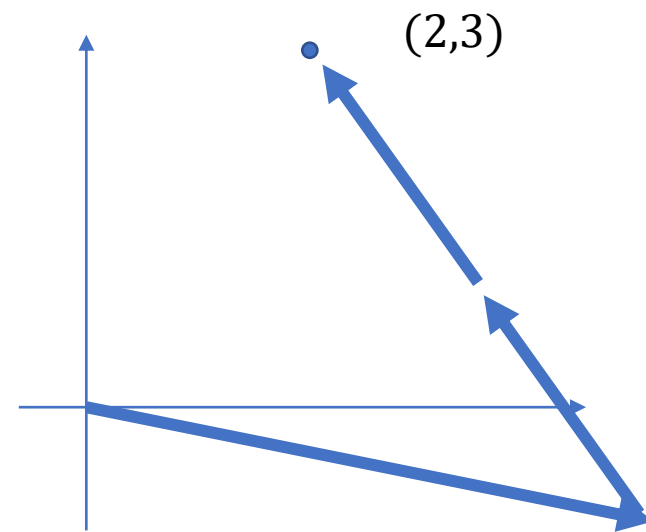
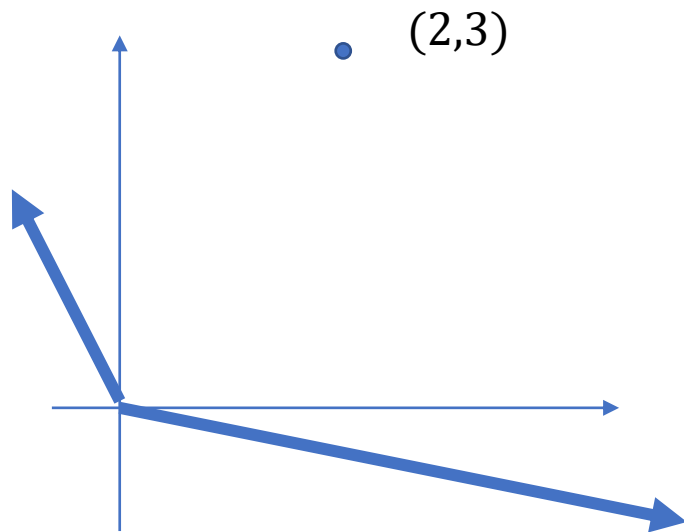
If $[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \dots \\ c_n \end{bmatrix}$, then the mapping $x \mapsto [x]_{\mathcal{B}}$ is the **coordinate mapping determined by $\mathcal{B}$**



We have seen, for example, that with basis $\mathcal{B} = \left\{ \begin{bmatrix} 4 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$, we have $[v]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$v \mapsto \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



Theorem 9

Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for vector space V . Then the coordinate mapping $x \mapsto [x]_{\mathcal{B}}$ is a one-to-one linear transformation from V onto R^n .

Proof: Show that this transformation preserves addition and scalar multiplication, then show it is one-to-one and onto.



Change of Coordinates Matrix

Given a basis $\mathcal{B} = \{b_1, \dots, b_n\}$ and a point $x = c_1 b_1 + \dots + c_n b_n$, we have

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \dots \\ c_n \end{bmatrix}$$

So if we define the matrix $P_{\mathcal{B}} = [b_1 \ b_2 \ \dots \ b_n]$ the equation is equivalent to

$$x = P_{\mathcal{B}} [x]_{\mathcal{B}}$$

$P_{\mathcal{B}}$ is called the **change of coordinates matrix** from $\mathcal{B}$ into the standard basis.



Change of Coordinates Matrix

Note that $P_{\mathcal{B}}$ is invertible and $P_{\mathcal{B}}^{-1}x = [x]_{\mathcal{B}}$.

$$x = P_{\mathcal{B}} [x]_{\mathcal{B}}$$

This matrix $P_{\mathcal{B}}^{-1}$ is the matrix associated with the linear transformation $x \mapsto [x]_{\mathcal{B}}$

A one-to-one linear transformation from a vector space V onto a vector space W is called an **isomorphism** from V onto W



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